

ally, vision system memory hierarchies can often be complex, distributed and not fully coherent.

Developers need a set of high-level tools and standard libraries like OpenCV and OpenVX that work in conjunction with and complement the underlying C/C++ tool chain. OpenCV is an open source computer-vision software library that contains 2500 functions which, when used by a high-level application, can facilitate tasks like object detection and tracking, image stitching, 3D reconstruction and machine learning.

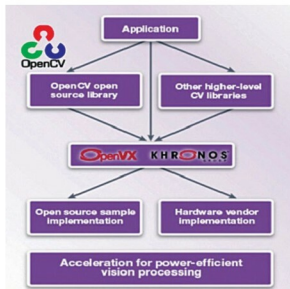


Fig. 8: OpenCV and OpenVX to accelerate vision system development

OpenVX contains a library of predefined and customisable vision functions, a graph-based execution model with task- and data-independent execution, and a set of memory objects that abstract the physical memory. It defines a 'C' application programming interface (API) for building, verifying and coordinating graph execution, as well as accessing memory objects.

Graph abstraction enables OpenVX implementers to optimise graph execution for the underlying acceleration architecture. It also defines the vxu utility library, which exposes each OpenVX predefined function as a directly callable 'C' function, without the need to first create a graph. Applications built using the vxu library do not benefit from optimisations enabled by graphs; however, the vxu library can be useful as the simplest way to use OpenVX and as the first step in porting existing vision applications. As computer-vision domain is still rapidly evolving, OpenVX provides an extensibility mechanism for adding developer-defined functions to the application graph.

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